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1 A General Periodic Fourier–State-Space Spatio–Temporal Model

1.1 Model setup

We consider a real-valued Gaussian spatio–temporal process

$$\{X_t(x) : x \in [0, 2\pi], t \in \mathbb{Z}\},$$

defined on a one-dimensional spatial domain with periodic boundary conditions (a circle) and discrete time. For each time t , the spatial field $X_t(\cdot)$ is represented by a truncated Fourier series

$$X_t(x) = a_0(t) + \sum_{j=1}^J \left[a_j(t) \cos(jx) + b_j(t) \sin(jx) \right], \quad x \in [0, 2\pi], \quad (1)$$

where J is a truncation level and

$$a_j(t), b_j(t), \quad j = 0, \dots, J, t \in \mathbb{Z},$$

are jointly Gaussian random coefficients. The representation (1) is exact in the limit $J \rightarrow \infty$ and is a finite-dimensional approximation in practice.

We collect the coefficients into a state vector

$$\mathbf{Z}_t = (a_0(t), a_1(t), \dots, a_J(t), b_1(t), \dots, b_J(t))^T \in \mathbb{R}^{2J+1}. \quad (2)$$

1.2 General linear Gaussian dynamics in Fourier space

The temporal evolution of the Fourier coefficients is specified by a linear Gaussian state-space model

$$\mathbf{Z}_{t+1} = \boldsymbol{\mu} + M (\mathbf{Z}_t - \boldsymbol{\mu}) + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \Sigma_\varepsilon), \quad (3)$$

where

- $\boldsymbol{\mu} \in \mathbb{R}^{2J+1}$ is the (possibly nonzero) mean state;
- $M \in \mathbb{R}^{(2J+1) \times (2J+1)}$ is the transition matrix;
- $\Sigma_\varepsilon \in \mathbb{R}^{(2J+1) \times (2J+1)}$ is the innovation covariance matrix;
- $\{\boldsymbol{\varepsilon}_t\}$ are i.i.d. Gaussian, independent of $\mathbf{Z}_0$.

Equation (3) implies the process $\mathbf{Z}_t$ is first-order Markov in time. The conditions for stationarity of $\{\mathbf{Z}_t\}$ are standard: if the spectral radius of M is strictly less than 1, i.e.

$$\rho(M) < 1,$$

then there exists a unique stationary distribution $\mathcal{N}(\boldsymbol{\mu}, \Sigma_0)$ for $\mathbf{Z}_t$, where Σ_0 solves the discrete-time Lyapunov equation

$$\Sigma_0 = M\Sigma_0M^\top + \Sigma_\varepsilon. \quad (4)$$

Given $\boldsymbol{\mu}$, M , and Σ_ε , the stationary covariance Σ_0 is determined by (4). Conversely, one can specify a desired stationary covariance Σ_0 and transition M , and then define

$$\Sigma_\varepsilon = \Sigma_0 - M\Sigma_0M^\top, \quad (5)$$

provided the right-hand side is positive semidefinite.

1.3 Matérn-like spatial margin on the circle

We seek to construct Σ_0 such that the induced spatial margin of $X_t(\cdot)$ at each time t is approximately that of a Matérn field on the circle.

In one spatial dimension on $\mathbb{R}$, the Matérn covariance with variance σ^2 , smoothness $\nu > 0$, and range parameter $\ell > 0$ can be written as

$$C_\nu(h) = \sigma^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\sqrt{2\nu} \frac{|h|}{\ell} \right)^\nu K_\nu \left(\sqrt{2\nu} \frac{|h|}{\ell} \right),$$

whose spectral density is proportional to

$$S(\omega) \propto (\kappa^2 + \omega^2)^{-(\nu+1/2)}, \quad \kappa = \frac{\sqrt{2\nu}}{\ell}.$$

On the periodic domain, we approximate this by imposing that the Fourier coefficients at a fixed time have variances

$$\sigma_j^2 \propto (\kappa^2 + j^2)^{-(\nu+1/2)}, \quad j = 0, \dots, J, \quad (6)$$

with $\kappa = \sqrt{2\nu}/\ell$. For simulation, it is convenient to normalize these so that

$$\sum_{j=0}^J \sigma_j^2 = \sigma_{\text{tot}}^2, \quad (7)$$

for a chosen total marginal variance σ_{tot}^2 . The choice of ν and ℓ controls, respectively, the spatial smoothness and range of $X_t(\cdot)$.

A minimal assumption consistent with (6) is that, at stationarity,

$$\text{Var}(a_j(t)) = \text{Var}(b_j(t)) = \sigma_j^2, \quad j = 1, \dots, J, \quad (8)$$

and $\text{Var}(a_0(t)) = \sigma_0^2$, with all these variances appearing along the diagonal of Σ_0 in (4). Cross-frequency and cosine/sine correlations can be incorporated in Σ_0 without changing the overall shape of the marginal spectral envelope given by (6).

1.4 Mode-specific temporal behaviour

The most general linear Gaussian model allows $\mathbf{Z}_t$ to evolve with a fully populated transition matrix M and arbitrary innovation covariance Σ_ε . For purposes of interpretation, it is useful to describe a structured, but still flexible, parameterization.

Frequency-dependent AR(1) persistence. For each frequency j , the coefficients may be assigned an effective persistence parameter $\rho_j \in (-1, 1)$ that governs their typical temporal autocorrelation length. In the simplest block-diagonal model (no cross-frequency coupling in M), one can set

$$a_j(t+1) = \rho_j a_j(t) + \eta_{a,j,t}, \quad b_j(t+1) = \rho_j b_j(t) + \eta_{b,j,t}, \quad j = 1, \dots, J, \quad (9)$$

and

$$a_0(t+1) = \rho_0 a_0(t) + \eta_{0,t}, \quad (10)$$

with Gaussian innovations

$$\eta_{0,t} \sim \mathcal{N}(0, \sigma_0^2(1 - \rho_0^2)), \quad \eta_{a,j,t}, \eta_{b,j,t} \sim \mathcal{N}(0, \sigma_j^2(1 - \rho_j^2)),$$

independent across j , between a and b , and across t . Then $\{a_j(t)\}$ and $\{b_j(t)\}$ are stationary AR(1) processes with variance σ_j^2 and autocorrelation $\rho_j^{|k|}$.

A key nonseparable feature arises from allowing the persistence to depend on frequency, e.g.

$$\rho_j = \exp(-\lambda j^\alpha), \quad j = 1, \dots, J, \quad (11)$$

with ρ_0 set separately in $(-1, 1)$, and parameters $\lambda > 0$ and $\alpha > 0$. In this case, low-frequency modes (small j) decorrelate slowly in time, while high-frequency modes decorrelate rapidly, so the effective spatial correlation structure changes with the time lag. This yields a space–time covariance that is stationary but generally nonseparable in space and time.

Cross-frequency and cosine/sine coupling. Beyond (9)–(10), one can introduce rich dependency structures in either

- the transition matrix M , allowing the value of mode j at time $t+1$ to depend on multiple modes at time t (e.g. banded or low-rank modifications of a block-diagonal structure), or
- the innovation covariance Σ_ε , allowing simultaneous shocks to be correlated across frequencies and between cosine/sine components.

For example, a block-structured M might have 2×2 blocks M_j acting on (a_j, b_j) for each j , with

$$M_j = \rho_j \begin{pmatrix} \cos \theta_j & -\sin \theta_j \\ \sin \theta_j & \cos \theta_j \end{pmatrix},$$

which induces a rotation (phase drift) and persistence at each frequency. Similarly, Σ_ε could be parameterized as

$$\Sigma_\varepsilon = \Lambda \Lambda^\top + D,$$

with a low-rank factor Λ representing a few latent shocks that affect many modes simultaneously, and D diagonal.

1.5 Space–time covariance

Under the simple independent-mode AR(1) structure (9)–(10) with zero mean, the space–time covariance

$$C(h, k) = \mathbb{E}[X_t(x)X_{t+k}(x+h)]$$

admits a closed form:

$$C(h, k) = \sum_{j=0}^J \sigma_j^2 \rho_j^{|k|} \cos(jh), \quad (12)$$

where $\cos(0 \cdot h) = 1$ for the $j = 0$ term. This follows by substituting (1), using the uncorrelatedness of different modes and cosine/sine components, and applying the identity

$$\cos A \cos B + \sin A \sin B = \cos(A - B).$$

Formula (12) makes explicit that the process is stationary in both space and time and that, unless all ρ_j are equal, $C(h, k)$ cannot be written as a separable product $C_{\text{space}}(h)C_{\text{time}}(k)$.

When M and Σ_ε are fully general, the covariance structure can still be characterized via the stationary covariance Σ_0 from (4), but the expression for $C(h, k)$ in physical space involves both frequency–lag-dependent auto- and cross-covariances among the Fourier coefficients and generally cannot be reduced to (12).

1.6 Parameter overview

The full model includes the following parameters, with varying levels of restriction depending on the chosen structure:

- **Fourier truncation:** J (integer, $J \geq 0$). Controls the finest spatial scale represented.
- **Matérn spatial parameters:**
 - $\nu > 0$ (smoothness): larger ν gives smoother spatial fields.
 - $\ell > 0$ (range): larger ℓ increases the spatial correlation length.
 - $\sigma_{\text{tot}}^2 > 0$ (total marginal variance): overall scale.
 - $\kappa = \sqrt{2\nu}/\ell$ (inverse range): internal scale parameter used in the spectral envelope (6).
- **Temporal persistence parameters:**
 - $\rho_0 \in (-1, 1)$: AR(1) persistence for the mean mode $a_0(t)$.
 - $\rho_j \in (-1, 1)$, $j = 1, \dots, J$, e.g. via (11) with
 - * $\lambda > 0$: controls overall decay speed across frequencies,

- * $\alpha > 0$: controls how strongly persistence changes with j .

- **State-space parameters:**

- $\boldsymbol{\mu} \in \mathbb{R}^{2J+1}$: mean state (for zero-mean processes, $\boldsymbol{\mu} = 0$).
- $M \in \mathbb{R}^{(2J+1) \times (2J+1)}$: transition matrix. Possible structures:
 - * block-diagonal in frequencies (independent modes),
 - * block-diagonal with 2×2 rotations per mode,
 - * banded, low-rank, or fully general.
- $\Sigma_\varepsilon \in \mathbb{R}^{(2J+1) \times (2J+1)}$: innovation covariance. Possible structures:
 - * diagonal (independent innovations),
 - * diagonal plus low-rank (factor structure),
 - * banded or fully general symmetric positive semidefinite.

- **Time horizon:** T (integer, $T \geq 1$). Number of time points simulated or observed.

By choosing simple structures (e.g. block-diagonal M with ρ_j given by (11), diagonal Σ_ε determined by (5), and $\boldsymbol{\mu} = 0$) one obtains a tractable, interpretable nonseparable model. More complex structures in M and Σ_ε can then be introduced gradually to capture richer cross-frequency interactions while preserving the Matérn-like spatial margin and Markovian temporal dynamics.